FogFarm

An IoT-Based Automated Aeroponic Cultivation System

Project Report

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1. Introduction

FogFarm is an automated aeroponic cultivation system that grows plants without soil, suspending roots in air inside an enclosed chamber and periodically misting them with a nutrient solution. The system combines a PVC-based grow structure with an ESP8266-based control unit that monitors environmental conditions and drives a misting pump automatically, removing the need for manual watering or environment monitoring.

1.1 Objectives

- Automate root-zone misting based on live humidity and temperature readings, without manual intervention.
- Maintain accurate day/night scheduling for growing lighting using a real-time clock (RTC).
- Log environmental data remotely for monitoring via ThingSpeak.
- Build a compact, safe, and presentable enclosure suitable for exhibition.
- Ensure reliable, low-heat, efficient power delivery to all onboard electronics.

2. System Overview

At its core, FogFarm is a closed-loop environmental control system. A DHT22 sensor continuously reads ambient humidity and temperature inside the grow chamber. When humidity drops below a set threshold or temperature rises above one, the ESP8266 microcontroller activates a relay-controlled misting pump for a fixed duration. A DS1307 real-time clock keeps track of time independently of network connectivity, driving a day/night lighting schedule. All sensor readings are periodically uploaded to ThingSpeak for remote logging and visualization.

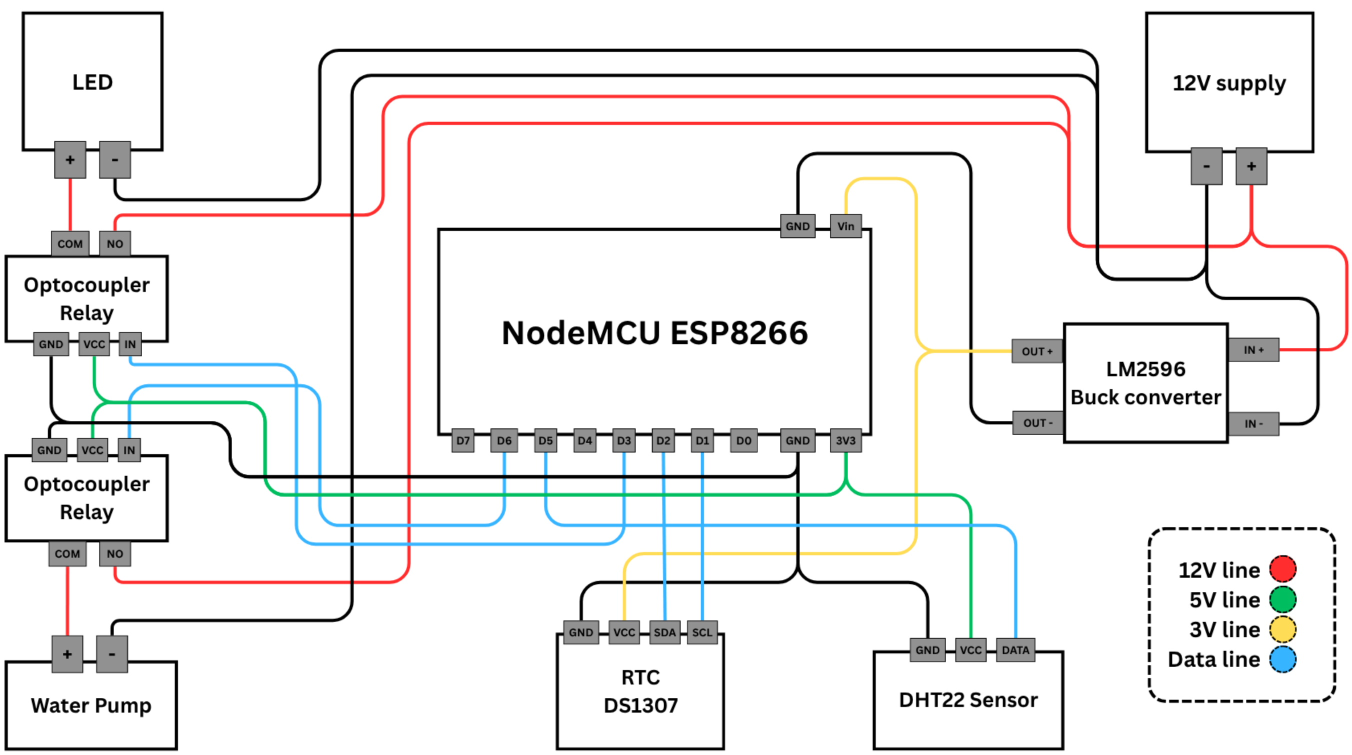
Subsystem	Function
Sensing	DHT22 measures humidity and temperature inside the chamber
Control logic	ESP8266 (NodeMCU) evaluates thresholds and drives the relay
Misting	Relay-controlled pump delivers a timed nutrient mist burst
Timekeeping	DS1307 RTC provides time for scheduling, independent of WiFi
Lighting	LED grow light switched on/off based on RTC-derived schedule
Data logging	ThingSpeak cloud channel receives humidity & temperature every 5min
Power	12V for pump and LED, 5V for esp and other setups
Structure	PVC frame enclosure with a covering layer

3. Hardware Components

Component	Role
ESP8266 (NodeMCU)	Main microcontroller; runs control logic, WiFi, ThingSpeak upload
DHT22	Digital temperature and humidity sensor
DS1307	I²C real-time clock with battery backup for scheduling
Optocoupler relay	Switches the misting pump and LED on/off
Diaphragm water pump	Delivers nutrient mist to suspended roots
LED grow light	Provides light on a day/night schedule
LM2596 Buck converter + 12V adapter	Steps 12V mains-adapter supply down to 5V logic rail
PVC pipe frame	Structural enclosure housing the root chamber and reservoir

3.1 Wiring Notes

- RTC: SDA → D2, SCL → D1 on the ESP8266.
- DHT22 data line: D5.
- Relay control: D6, wired active-LOW (LOW = pump ON).
- LED control: D3.



4. Firmware Logic

The ESP8266 firmware runs a control loop that performs, in sequence: RTC time read, hourly pump-cycle counter reset, DHT22 sensor read, misting decision, relay timing, ThingSpeak upload check, and light scheduling.

4.1 Misting Control

A misting cycle is triggered when humidity falls below threshold or temperature rises above threshold, the cooldown period has elapsed since the last trigger, and the hourly cycle limit has not been reached. This prevents over-misting and protects the pump from excessive duty cycling.

Parameter	Purpose	Typical value
Humidity threshold	Minimum humidity before misting triggers	90%
Temperature threshold	Maximum temperature before misting triggers	27°C
Relay duration	How long the pump stays on per cycle	5s
Cooldown	Minimum wait between pump activations	20 min
Max cycles/hour	Hard cap on pump activations per hour	3

4.2 Scheduling & Data Logging

- Grow light: ON between 6:00 and 22:00 based on RTC hour, OFF otherwise.
- ThingSpeak upload: every 15 seconds, humidity and temperature are posted to the ThingSpeak channel over a raw HTTP request via WiFiClient.
- The lamp indicator is also updated in ThinkSpeak.

5. Power Supply:

12V 5A adapter has been used along with buck converter at 5V output for powering the system and

Component	Power Source
Esp8266	5V (Buck converter)
Diaphragm water pump	12V DC mains adapter
LED	12V DC mains adapter
Relays	5V (Buck converter)
Sensors and other components	5V (Buck converter)

6. Real-Time Clock Accuracy

The DS1307 RTC uses a standard, non-temperature-compensated 32.768 kHz crystal, which is subject to drift on the order of seconds to minutes over days or weeks depending on ambient temperature. Minor drift (observed: ~6 seconds early on a scheduled light-off event) has been noted and is expected behaviour for this class of RTC. A future revision may substitute a DS3231 (temperature-compensated RTC) for improved long-term accuracy, or reintroduce periodic NTP re-synchronization over WiFi.

7. Cost analysis

Expenditure	Amount
Structural Cost	₹2811
Electrical Component Cost	₹2657
Material Losses	₹532
Total	₹6000

8. Conclusion

FogFarm demonstrates a functional, low-cost aeroponic cultivation system that automates the core tasks of soil-free plant growth: root-zone misting, day/night lighting, and environmental monitoring. By combining a DHT22 sensor, an ESP8266 control unit, and a DS1307 RTC, the system maintains stable humidity and temperature conditions inside the grow chamber without requiring manual intervention, while remote logging to ThingSpeak allows conditions to be tracked and verified in real time.

Testing confirmed that the misting logic — governed by humidity/temperature thresholds, a cooldown period, and an hourly cycle cap — reliably prevents both under- and over-misting, protecting the pump from excessive duty cycling. The RTC-driven lighting schedule performed as expected, with only minor timekeeping drift (on the order of a few seconds) attributable to the non-temperature-compensated crystal used in the DS1307, a known and expected limitation of this class of RTC.

Overall, the system met its stated objectives: automated environmental response, independent scheduling, remote data logging, and a compact, presentable enclosure suitable for exhibition. Power delivery via the buck-converted 5V rail alongside a direct 12V rail for the pump and LED proved stable and efficient throughout testing.